

HALKIRK AGCM, AB, Canada

WMO: 717920

Lat: 52.1166N

Lon: 112.1641W

Elev: 2766

StdP: 13.29

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-23.0	-18.3	-29.0	1.2	-22.6	-24.3	1.6	-18.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.5	84.5	61.5	80.9	60.6	77.5	59.5	65.3	76.8	63.3	76.1	61.6	73.8	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.2	89.5	70.5	58.9	82.4	67.8	56.9	76.5	65.7	31.7	76.5	30.1	76.0	28.8	74.7	73.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-31.3	90.4	4.4	4.7	-34.5	93.8	-37.1	96.5	-39.6	99.2	-42.9	102.6
			-30.4	69.3	5.7	2.8	-34.4	71.3	-37.8	72.9	-41.0	74.4	-45.1	76.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	14.1	11.6	23.5	37.0	50.3	57.4	62.1	60.6	51.7	38.8	23.8	11.7		
	DBStd	21.50	14.31	14.94	14.24	9.75	7.86	5.56	5.10	5.98	8.24	9.16	13.05	14.09		
	HDD50	5928	1114	1075	823	398	94	9	0	3	77	362	786	1187		
	HDD65	10275	1579	1495	1288	840	458	234	118	161	402	813	1235	1652		
	CDD50	1192	0	0	0	8	103	231	375	332	128	14	1	0		
	CDD65	63	0	0	0	0	1	6	28	25	3	0	0	0		
	CDH74	1591	0	0	0	6	93	155	575	572	185	5	0	0		
	CDH80	401	0	0	0	0	10	24	159	163	45	0	0	0		
	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	PrecAvg	16.7	0.8	0.5	0.7	1.2	1.9	3.1	3.0	2.2	1.3	0.7	0.6	0.6		
(15) Precipitation	PrecMax	20.8	2.5	1.0	1.5	2.4	3.3	4.5	6.5	4.2	3.0	1.8	2.2	1.2		
	PrecMin	11.4	0.0	0.1	0.1	0.2	0.6	1.0	0.9	0.2	0.1	0.2	0.1	0.2		
	PrecStd	2.8	0.6	0.3	0.4	0.6	0.8	1.0	1.3	0.9	0.8	0.4	0.5	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.5	43.6	56.9	71.1	81.2	83.2	88.7	89.1	85.5	72.9	58.6	44.4		
		MCWB	37.2	37.2	44.9	49.4	55.6	60.8	64.4	62.6	59.0	53.9	45.3	37.8		
	2%	DB	40.4	38.3	50.2	65.0	76.4	78.3	84.5	84.2	79.2	64.5	51.1	39.0		
		MCWB	35.4	33.3	40.8	47.7	53.7	59.4	62.9	61.4	57.1	49.6	41.7	33.7		
	5%	DB	37.7	35.0	44.8	59.1	72.0	74.6	80.5	80.5	74.4	59.1	44.2	34.7		
		MCWB	33.7	31.3	37.9	43.9	52.3	58.0	62.7	61.0	55.8	46.6	37.4	30.6		
	10%	DB	33.8	31.6	40.1	53.3	67.1	70.9	76.4	76.3	68.6	53.5	39.9	30.8		
		MCWB	30.1	29.2	34.9	41.1	50.4	56.4	61.6	60.0	53.6	43.9	34.5	27.7		
	0.4%	WB	37.6	37.5	46.0	51.7	57.6	65.0	69.7	67.3	61.4	55.1	46.4	38.4		
		MCDB	42.5	42.9	56.1	69.1	76.0	76.7	81.1	77.8	79.0	71.0	58.0	43.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	35.7	33.7	41.2	47.8	55.3	61.7	66.0	65.0	58.7	50.9	41.9	34.1		
		MCDB	39.9	37.6	49.1	63.3	71.2	73.7	77.0	76.7	75.8	61.1	50.0	38.8		
	5%	WB	33.8	31.7	38.1	44.8	53.6	59.7	63.8	62.9	56.3	47.6	38.0	30.9		
		MCDB	37.8	34.7	44.4	57.2	69.3	70.6	76.3	76.6	71.2	57.6	43.2	34.3		
	10%	WB	30.5	29.2	35.1	42.1	51.7	57.8	62.3	60.9	54.4	44.5	35.0	27.8		
		MCDB	33.7	31.8	39.5	52.1	64.4	68.3	74.6	73.2	68.2	52.9	38.9	30.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	17.4	18.1	18.4	21.4	24.1	22.0	24.5	25.3	24.7	21.5	17.6	16.7		
		MCDBR	17.8	19.0	23.1	32.4	31.9	28.6	32.5	34.0	35.2	30.1	23.1	19.1		
		MCWBR	14.0	15.5	14.7	16.8	14.0	12.8	14.2	14.7	16.1	16.8	14.7	15.4		
	5% WB	MCDBR	18.5	18.8	22.0	29.6	27.6	25.1	28.3	28.5	32.2	28.3	21.4	19.2		
		MCWBR	14.9	15.7	14.5	16.0	12.7	12.7	14.0	13.9	15.5	16.2	14.3	15.8		
(40) Clear-Sky Solar Irradiance	taub		0.267	0.283	0.285	0.328	0.353	0.358	0.352	0.366	0.315	0.283	0.256	0.246		
	taud		2.293	2.336	2.451	2.367	2.343	2.367	2.418	2.358	2.467	2.537	2.471	2.339		
	Ebn at Noon		235	267	292	289	285	284	284	272	277	268	246	227		
	Edh at Noon		20	25	28	34	37	36	34	35	27	21	16	16		
(44) All-Sky Solar Radiation	RadAvg		349	681	1127	1507	1830	1925	2037	1658	1164	698	362	263		
	RadStd		30	56	87	134	119	122	101	120	121	85	34	25		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (80 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page